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Cc:
Subject: [External] AI Action Plan
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Scalable, Resource-Efficient AI Infrastructure – LILA Net & Decentralized AI Compute

To the National Science Foundation AI Policy Committee,

The U.S. must lead AI development through sustainable, scalable, and transparent infrastructure that reduces energy strain, improves AI accountability, and enhances national security.

Challenges Facing Current AI Infrastructure:

1. Excessive compute demands that strain power grids and limit AI scalability.
2. Lack of transparency & oversight, increasing risk in high-impact AI applications.
3. Inefficient infrastructure models that rely on centralized data centers, making AI deployment costly and resource-intensive.

Proposed Solution: LILA Net & Decentralized AI Compute

1. LILA Net – AI-to-AI Communication & Efficient Compute Scaling

LILA (Lightweight Integrated Language Algorithm) is a distributed AI framework designed to:

- Reduce compute burden by decentralizing AI processing across modular node clusters.
- Improve efficiency through Multi-AI GAN training, where AI models collaboratively refine each other, reducing training time by up to 30% compared to single-agent training.
- Enhance transparency with built-in audit logging & oversight mechanisms for real-time

accountability.

- Reduce AI-to-AI communication latency by an estimated 40%, enabling faster model coordination and real-time decision-making.

2. Decentralized AI Compute – Scaling Without Straining National Infrastructure

To prevent AI models from overloading public data centers, we propose using clustered smart compute nodes (e.g., NVIDIA DIGITS or similar U.S.-manufactured devices), which:

- Reduce energy consumption by 40-60% compared to traditional server stacks, thanks to specialized power-efficient AI hardware.
- Eliminate single points of failure, making AI more resilient and adaptive.
- Enable modular scaling, meaning additional AI nodes can be deployed without requiring new data center infrastructure.

Why This Matters: Unlike traditional AI deployments that depend on massive centralized data centers, modular node clusters distribute processing closer to the point of use, making AI faster, more efficient, and scalable without major energy costs.

Policy Recommendations for the National AI Action Plan

1. Allocate an initial \$10 million in funding for:

- Pilot deployments of LILA Net in strategic AI hubs (government, healthcare, defense).
- Development of standardized AI-to-AI communication protocols to enhance model interoperability.
- Optimization research on decentralized AI hardware configurations to ensure scalability and efficiency.

2. Develop national standards for AI-to-AI communication to ensure transparent, scalable, and accountable AI operations.

3. Provide targeted tax incentives for AI companies implementing low-energy AI computing solutions that align with NSF sustainability goals.

4. Refined Data Ecosystem – Securing U.S. Data Sovereignty & Responsible AI Use

A robust data governance framework is essential for realizing the full potential of LILA Net and decentralized AI compute, ensuring responsible AI development and protecting U.S. data

sovereignty.

Key Components of a U.S.-Controlled Data Ecosystem:

U.S. Data Sovereignty & Controlled Access

- Mandate U.S.-based hosting & processing for AI models using American citizen data.
- Restrict foreign access to critical-sector datasets (finance, defense, healthcare).
- Establish federal AI data hubs for secure government-managed datasets.

Granular User Control & Privacy

- User-controlled consent frameworks for managing personal data in AI training.
- Decentralized identity verification systems ensure privacy while keeping AI interactions traceable.
- Real-time data anonymization protects users while enabling AI-driven personalization.

Organizational Data Protection

- AI models interacting with company data should operate under a zero-trust architecture.
- Federated learning protocols enable AI training on distributed datasets without sharing raw data.
- Differential privacy techniques ensure aggregated enterprise data cannot be reverse-engineered.

Challenges & Mitigation Strategies

- Technical Complexity → Funding for AI privacy research & development will ensure scalable, secure deployment.
- Balancing Innovation & Privacy → Federated learning & encryption ensure AI insights without compromising sensitive data.

Policy Recommendations:

1. Mandate U.S.-based AI data hosting to prevent foreign access risks.
2. Implement user-level privacy controls to enhance data security & informed consent.
3. Require zero-trust frameworks for AI models handling corporate & government data.
4. Launch a federally backed National AI Data Infrastructure Initiative to establish U.S. leadership in AI governance.

Conclusion

By integrating LILA Net, decentralized AI compute, and a robust data governance framework, the U.S. can address the critical challenges of scalability, transparency, and security in AI development, ensuring American leadership in this transformative technology.

This approach will:

Prevent AI compute bottlenecks while reducing energy grid strain.

Increase AI transparency & security with auditable AI-to-AI communication.

Enable cost-effective innovation without reliance on centralized data centers.

By implementing these strategic AI infrastructure solutions, the United States will maintain technological leadership while ensuring responsible, transparent, and secure AI development for the future.

Thank you for your consideration.

Sincerely,

Thomas Giza

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requirements.

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